LING285-L7

Course: LING285: Human Language and Technology

Title of Lecture: Lecture 7: Dynamical Systems 5

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Section 1: The Harmonic Oscillator

- A **harmonic oscillator** is a dynamical system with two variables that feed into each other: **position**, written x, and **velocity**, written v. Position tells us where something is, and velocity tells us how fast and in which direction it is moving. The system is fully described by two rules of change, one for each variable, and each rule depends on the other variable.
    - The dot above a letter is the notation for change over time. $\dot{x}$ means the **increase or decrease in x with time**, so it is the rate at which position is changing at a given moment.
    - In the same way, $\dot{v}$ means the **increase or decrease in v with time**, so it is the rate at which velocity is changing.
- The first rule says that the rate of change of position is equal to the velocity:

$$\dot{x} = v$$

- This rule says that velocity drives position. When v is positive, x grows, and when v is negative, x shrinks. When v is exactly zero, x holds still for that instant.
- In the loop diagram, this is the red arrow running from v to x, which marks v as the cause and x as the thing being changed.
- The second rule says that the rate of change of velocity is equal to the negative of the position:

$$\dot{v} = -x$$

- This rule says that position drives velocity, but with a minus sign. When x is positive, velocity is pushed downward, and when x is negative, velocity is pushed upward.
- This is the blue arrow running from x to v. Together, the red and blue arrows form a **closed loop of mutual influence**: v changes x, and x changes v, over and over.
- We can work out what this loop does by stepping through it by hand, starting from $x = 1$ and $v = 0$. At the very start, position is at its highest value and nothing is moving yet.
    - Since $x = 1$, the second rule gives $\dot{v} = -1$, so velocity starts decreasing and becomes negative.
    - Once v is negative, the first rule says $\dot{x}$ is negative, so position begins to fall. It falls slowly at first, since v has only just started to become negative, and then falls faster and faster as v grows more negative.
    - Position keeps falling until it crosses zero. At the moment $x = 0$, the second rule gives $\dot{v} = 0$, so velocity stops changing. This is the moment when velocity is at its most negative, which means position is falling at its fastest right as it passes through zero.
    - After the crossing, x is negative, so $\dot{v} = -x$ is now positive. Velocity starts climbing back toward zero, which means position is still falling but falling more slowly.
    - When velocity reaches zero, position stops falling. This is the lowest point of x, at -1. From there, velocity turns positive, position starts to rise, and the whole sequence runs in reverse until x is back at $+1$ with $v = 0$.
- Plotting both variables against time point by point gives two waves. Position starts at its peak of $+1$, drops through zero, bottoms out at -1, and comes back up to its peak.

Velocity starts at zero, dips negative, returns to zero, rises positive, and comes back to zero.
    - The two waves are offset from each other by a quarter of a cycle. **Velocity is zero whenever position is at a peak or a trough**, and **velocity is at its largest size whenever position is crossing zero**.
    - This offset is a direct consequence of the two rules. Position turns around only when velocity reaches zero, and velocity turns around only when position reaches zero.

Section 2: The Oscillation/Vibration Dynamical System

- An **oscillation**, also called a **vibration**, is a repeating back-and-forth motion around a central point. The two-variable loop between x and v is the basic dynamical system that produces it. Each variable pushes on the other, and the result is that neither one ever settles down.
    - When the loop is drawn with one arrow from x to v and another from v back to x, it shows that the influence is **circular**. There is no single cause at the start and no single effect at the end.
    - Graphed smoothly over time, x traces a wave that starts high, swings down below zero, swings back up, and continues this pattern through repeated cycles. v traces a wave of the same shape that starts at zero and heads negative first, staying a quarter cycle out of step with x.
- The key property of this system is that **it never comes to rest on its own**. In the systems from earlier dynamical-systems material, variables moved toward a resting value and stopped there. Here, every time one variable reaches its target, the other variable is at its largest value and pushes it past the target again.
    - This is why the motion keeps repeating. Position overshoots zero because velocity is at its maximum at the moment position reaches zero.
    - Velocity overshoots zero because position is at its maximum at the moment velocity reaches zero.

Section 3: Vibration Needs Two Physical Effects

- Physically, vibration requires **two effects working together**. Neither one alone can produce a back-and-forth motion.
- The first effect is a **spring**, more precisely a **viscous spring**. A spring is the springiness found in many objects around us, meaning their tendency to return to a natural shape or position after being pushed or pulled. The word "viscous" refers to a frictional, slowing quality, and it will be set aside here so that we can focus on the springiness itself.
    - A spring has a property called **stiffness**. Stiffness measures how strongly the spring pulls back toward its natural position when it has been moved away from it.
    - A very stiff spring pulls back hard, and a loose spring pulls back gently.
- The second effect is **mass**. Mass is the amount of matter in an object, and it determines how the object carries motion once it is moving.
- Vibration matters because it shows up across many different systems, both physical and abstract.
    - **The jolt we feel when a car hits a bump** is a vibration. The car's suspension springs and the car's mass together produce a bouncing motion.
    - **Music** is vibration. Strings, air columns, and drum skins all vibrate to produce sound.
    - **Boom and bust in economics** is a vibration. An economy swings above and below a steady level in repeating cycles.
    - **Oscillation in SGD** is a vibration. SGD stands for **stochastic gradient descent**, a method used to train machine-learning models by repeatedly nudging a value toward a target. Under some settings, that value swings back and forth around the target rather than settling onto it.
    - **The motion of the air in the vocal tract and the motion of the vocal folds during speech** are vibrations. Every sound we produce when we talk is built from these vibrations.

Section 4: A Slice of Air in the Vocal Tract

- The air inside the vocal tract can be treated as a small slice of material made of many particles. During speech, the particles in the slice stay the same, and each particle stays the same size. **What changes is the distance between the particles.**
  - At **equilibrium spacing**, the particles sit at their natural, resting distance from one another. A chosen pair of neighboring particles is at **100% of its equilibrium distance** at this point.
  - The equilibrium size of the slice serves as the reference that all other states are compared to.
- **Compression** is the state in which the particles are pushed closer together than their resting distance. In a compressed slice, the chosen pair of particles sits at only **56% of its equilibrium distance**, and the whole slice shrinks.
  - **High pressure means air that is packed very close together.** Compression is therefore a high-pressure state.
- **Rarefaction** is the state in which the particles are spread farther apart than their resting distance. In a rarefied slice, the chosen pair sits at **144% of its equilibrium distance**, and the whole slice expands.
  - **Low pressure means air that is spread very far apart.** Rarefaction is therefore a low-pressure state.
- During speech, a slice of air in the vocal tract alternates between compression and rarefaction, shrinking and expanding around its equilibrium size again and again. This alternation is a vibration of the air.
  - The sizes used in this model are idealized and exaggerated so that the change in spacing is visible. Real air in the vocal tract changes its spacing by much smaller amounts.

Section 5: The Viscous Spring and Negative/Balancing Feedback

- The spring effect on its own can be written as a single rule of change:

$$\dot{x} = -\frac{1}{3}x$$

- Here x is the distance of the spring from its natural position, and $\dot{x}$ is how fast that distance is changing. The rule says the rate of change is always **one third of the current distance, with the opposite sign**.
- The sign of x decides which way the spring moves.
  - **If x is positive, $\dot{x}$ is negative, so x falls.** A spring stretched in the positive direction moves back toward zero.
  - **If x is negative, $\dot{x}$ is positive, so x rises.** A spring compressed in the negative direction also moves back toward zero, just from the other side.
- We can follow the rule step by step starting from an **initial condition** of $x = 1$. The initial condition is the value the system starts from before any change has happened.
  - With $x = 1$, the rule says x will fall, and it will **fall by one third of its current size.** One third of 1 is $\frac{1}{3}$, so x drops to:

$$1 - \frac{1}{3} = \frac{2}{3}$$

- The same rule then applies again to the new value. One third of $\frac{2}{3}$ is $\frac{2}{9}$, so x drops to:

$$\frac{2}{3} - \frac{2}{9} = \frac{4}{9}$$

- Applying the rule once more, one third of $\frac{4}{9}$ is $\frac{4}{27}$, so x drops to:

$$\frac{4}{9} - \frac{4}{27} = \frac{8}{27}$$

- The process repeats **again and again**, and each drop is smaller than the one before it. The first drop was $\frac{1}{3}$, the second was $\frac{2}{9}$, and the third was $\frac{4}{27}$.

- **When x is smaller, $\dot{x}$, its speed of falling, is smaller too.** The spring moves quickly when it is far from its natural position and slows down as it gets closer.
    - When graphed against time, x starts at 1 and curves downward toward zero. The slope of the curve is steeply negative at the start and becomes **smaller and smaller negative** as time goes on.
    - The curve gets closer and closer to zero but **never crosses it**. There is nothing in this rule that could carry x past zero, since the speed shrinks to nothing as the distance shrinks to nothing.
- If the initial condition is negative, the same logic runs in the opposite direction. Starting below zero, x rises toward zero, quickly at first and then more and more slowly, with a slope that is **smaller and smaller positive**. It approaches zero from below and again never crosses it.
- This kind of system has a **natural resting condition**, which here is $x = 0$**.** When the system's position is at a **discrepancy** with its natural position, meaning it has been moved away from where it rests, it **feeds back on itself** through the rule $\dot{x} = -\frac{1}{3}x$ to cancel that discrepancy. This is the idea of **negative feedback**, also called **balancing feedback**, from Meadows' work on systems.
    - The feedback is called negative because the change always opposes the discrepancy. A positive discrepancy produces a negative change, and a negative discrepancy produces a positive change.
    - It is called balancing because its effect is to bring the system back into balance at its resting point.
- **These are systems that are in motion only in order to achieve their equilibrium.** An **equilibrium** is a state where the system has no reason to change any further. A spring moves only because it is away from its natural position, and once it arrives there, all motion stops.
- The number $-\frac{1}{3}$ in the rule is what we call **stiffness**. Its size controls how quickly the spring returns to equilibrium.
    - A **higher stiffness**, such as a larger fraction than one third, means the spring closes a bigger share of the remaining distance at every step, so it returns to zero faster.
    - A **lower stiffness**, such as a smaller fraction than one third, means the spring closes a smaller share of the remaining distance at every step, so it returns to zero more slowly.

Section 6: Mass and Persistent Motion

- Mass behaves very differently from a spring. **If a mass is moving, it just keeps on moving.** A mass does not need a discrepancy or a target to keep going, and it has no natural resting position that it is trying to reach.
- This principle was worked out by **Galileo, Descartes, and Newton**: **a mass in motion stays in motion**. Without an outside force acting on it, a moving object continues at the same speed in the same direction.
    - Motion of this kind is called **constant velocity**. An object with constant velocity covers **the same distance in every equal interval of time**.
    - A car rolling along a straight street past buildings and trees at constant velocity moves past the same amount of scenery in every moment. It does not speed up, slow down, or stop on its own.
- This property is exactly what the spring lacks. A spring slows down as it nears its resting point and stops there, while a mass carries its motion forward regardless of where it is.

Section 7: A Spring Stops at a Place, a Mass Stops After a Process

- The difference between the two effects can be stated in one line: **a spring stops at a place, while a mass stops after a process**. The spring's stopping point is a location, its natural position. A mass does not stop at a location at all, and it keeps moving until something removes its motion over time.
- Consider a spring attached to a wall on one end, lying on a table, with no mass on its free end. Its displacement x is **measured from the spring's natural position**, which is set as $x = 0$, and the scale runs from -1 to $+1$.

    - When the free end is pulled out to $+1$ and released, the spring **returns and stops at** $x = 0$. The end approaches zero **without crossing it**.
    - Graphed over time, this return from $+1$ is a curve that starts at $+1$ and bends down toward zero, flattening out as it gets close. It has the same shape as the curve produced by $\dot{x} = -\frac{1}{3}x$.
    - When the free end is instead pushed to -1 and released, x **rises toward zero without crossing it**. The curve starts at -1 and bends up toward zero, again flattening as it arrives.
    - Pulling the spring out again gives the exact same return each time. A spring alone always settles at its natural position and never overshoots.
- When a **mass** is attached to the free end of the same spring and the system is released from $+1$, the result is completely different. The spring still pulls the mass back toward zero, but when the mass reaches zero it is moving at its fastest, and **the mass carries that motion through the crossing**.
    - The mass continues past zero into negative displacement. Now the spring is stretched the other way, so it pulls back toward zero from the negative side. The mass slows down, stops near -1, and is pulled back through zero toward the positive side again.
    - This repeats through **every crossing**. The mass keeps going back and forth past the natural position rather than stopping at it.
- Graphed over time, the spring-plus-mass system produces a **lightly damped oscillation**. The curve swings from $+1$ down below zero, back up above zero, and continues swinging through repeated cycles.
    - **Damping** is the gradual loss of motion over time, which comes from the viscous, frictional part of the spring. In a lightly damped system, the damping is small, so each swing is only a little smaller than the one before it and the oscillation continues for many cycles.
    - Eventually, the damping would bring the mass to rest at zero, which is what it means for a mass to stop after a process. The stopping happens only at the end of a long sequence of swings.
- Putting the two effects together recovers the harmonic oscillator. The spring supplies the pull back toward the natural position, which is the rule $\dot{v} = -x$: the farther the mass is from zero, the harder its velocity is pushed back toward zero. The mass supplies persistent motion, which is the rule $\dot{x} = v$: position keeps changing as long as there is velocity, including at the moment the mass passes through zero.
    - A spring alone gives balancing feedback that ends at equilibrium.
    - A mass alone gives motion that never turns around.
    - **A spring and a mass together give vibration**, since the spring keeps turning the motion around and the mass keeps carrying it past the resting point.

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Section 8: A Cosine in the Vocal Tract

- The vibration of a slice of air in the vocal tract can be described with a single mathematical function, the **cosine**. The value x here measures the **normalized expansion** of the slice, meaning how far the particles are spread relative to their equilibrium spacing, scaled so that the extremes are $+1$ and -1.

$$x = \cos(\text{phase})$$

- $x = +1$ means the particles are **farthest apart**, which is **maximum rarefaction**.
- $x = 0$ means the particles are at their **equilibrium** spacing.
- $x = -1$ means the particles are **closest together**, which is **maximum compression**.
- The **phase** is the input to the cosine, and it tracks where the slice is within its cycle of motion. One full cycle of the cosine takes a phase of 2π, so a phase running from 0 to 6π covers exactly **three cycles**.
    - At a phase of 0, the cosine equals $+1$, so the slice starts at maximum rarefaction.

    - At a phase of π, the cosine equals -1, so the slice has reached maximum compression halfway through the first cycle.
    - At a phase of 2π, the cosine is back at $+1$, so the slice has returned to maximum rarefaction and completed Cycle 1. The same pattern repeats for Cycle 2, ending at 4π, and Cycle 3, ending at 6π.
- The slice therefore moves from **maximum rarefaction to compression and back to rarefaction** in every cycle, following the cosine curve exactly.
    - Partway through a swing, the slice is in an in-between state. At $x = +0.84$, the particles are still rarefied, farther apart than at equilibrium but not at their maximum spread. At $x = -0.84$, the particles are compressed, closer together than at equilibrium but not at their tightest.
- This cosine is the same wave that position traced in the harmonic oscillator, starting at its peak and swinging down through zero to its lowest point and back. The air in the vocal tract behaves as a harmonic oscillator, with the springiness of the air pushing it back toward equilibrium spacing and the mass of the air carrying it past equilibrium each time.
    - The sizes of the slice in this model are exaggerated and its motion is slowed, so that each stage of the cycle is visible. The dashed boundary marks the slice's equilibrium size.

Section 9: Vowels and Tongue Position

- Different vowels are made by placing the tongue in different positions inside the vocal tract. Side-view MRI images of the head during five words show how the tongue body moves to create each vowel. The vowels are organized along two dimensions: **height**, which is how high or low the tongue body sits in the mouth, and **front/back**, which is how far forward or backward the tongue body sits.
- **High vowels** are made with the tongue body raised close to the roof of the mouth.
    - In **"dean,"** the vowel is **high and front**. The tongue body is pushed up and forward toward the hard palate, which leaves a wide open space in the throat behind the tongue.
    - In **"dune,"** the vowel is **high and back**. The tongue body is raised but bunched toward the back of the mouth, closer to the soft palate.
- **Low vowels** are made with the tongue body lowered away from the roof of the mouth.
    - In **"Dan,"** the vowel is **low and front**. The tongue sits lower in the mouth while staying toward the front, leaving a large open space above it.
    - In **"Don,"** the vowel is **low and back**. The tongue is lowered and pulled back toward the throat, which narrows the space behind the tongue.
- In **"done,"** the vowel is **central**. The tongue body sits in a middle position, neither pushed far forward nor pulled far back.
- Each tongue position gives the air inside the vocal tract a different shape. A different shape changes how the air vibrates, and those differences in vibration are what make one vowel sound different from another.

Section 10: One Time Interval, Three Frequencies

- The **frequency** of a vibration is how many complete cycles it goes through in a given amount of time. Three vibrations can be compared over the same 12-second interval, with each one covering the same range of particle spacing and each one starting at **maximum rarefaction**, where $x = +1$.
    - $x = +1$ is maximum rarefaction, $x = 0$ is equilibrium, and $x = -1$ is maximum compression, just as in the single cosine.
- The **low-frequency** vibration completes **3 cycles in 12 seconds**. Each cycle takes 4 seconds, so its peaks at $+1$ fall at 0, 4, 8, and 12 seconds. Measured in cycles per second, its frequency is:

$$\frac{3 \text{ cycles}}{12 \text{ s}} = 0.25 \text{ cycles per second}$$

- The **mid-frequency** vibration completes **6 cycles in 12 seconds**. Each cycle takes 2 seconds, so the slice swings between rarefaction and compression twice as often as the low-frequency one. Its frequency is:

$$\frac{6 \text{ cycles}}{12 \text{ s}} = 0.5 \text{ cycles per second}$$

- The **high-frequency** vibration completes **9 cycles in 12 seconds**. Each cycle takes $\frac{12}{9}$, or about 1.33 seconds, so this slice swings the most rapidly of the three. Its frequency is:

$$\frac{9 \text{ cycles}}{12 \text{ s}} = 0.75 \text{ cycles per second}$$

- All three vibrations reach the same extremes of rarefaction and compression. **The only thing that differs among them is how quickly the cycle repeats.** Frequency is therefore a separate property from how far the particles spread or squeeze, and a vibration can be made faster or slower without changing its range.
    - One cycle per second is called **one hertz**, written **Hz**. This unit is used for the frequencies in the vowel models.

Section 11: Vowels as Combined Vibrations

- A vowel is not a single cosine. A vowel can be modeled as **a sum of vibrations at different frequencies happening at the same time** in the same slice of air. The simplest model uses two frequencies of equal size added together, and the two frequencies chosen give each vowel its own pattern of motion.
    - The frequencies in these models are small illustrative values, chosen so the combined wave is easy to follow. They are not the actual frequencies found in speech.
- A cosine at a frequency of f Hz can be written as $\cos(2\pi f t)$, where t is time in seconds. Since one cycle takes a phase of 2π, multiplying 2π by f gives f full cycles for every second that passes.
    - A 1 Hz vibration is $\cos(2\pi t)$, which reaches a phase of 2π, or one full cycle, at $t = 1$ second.
    - A 4 Hz vibration is $\cos(8\pi t)$, which reaches a phase of 8π, or four full cycles, at $t = 1$ second.
- The vowel /**i**/, the vowel in "dean," is modeled as a **1 Hz vibration plus a 4 Hz vibration**:

$$x(t) = \cos(2\pi t) + \cos(8\pi t)$$

- At $t = 0$, both cosines equal $+1$, so the sum is:

$$x(0) = 1 + 1 = 2$$

- This means the combined wave starts at $+2$, which is **maximum rarefaction** for the combined motion. The range of x is now from $+2$ to -2, since two waves of size 1 are stacked together.
- At $t = 0.5$ seconds, the 1 Hz wave is halfway through its cycle and the 4 Hz wave has completed exactly two cycles, so the sum is:

$$x(0.5) = \cos(\pi) + \cos(4\pi) = -1 + 1 = 0$$

- Between its large peaks, the /i/ wave has a busy pattern: it rises to smaller peaks near $+1$, dips down to troughs near -1.7, and touches zero at the halfway point. The fast 4 Hz wave rides on top of the slow 1 Hz wave, putting several smaller wiggles inside each slow cycle.
- The whole pattern repeats **once every second**, returning to $+2$ at 1, 2, and 3 seconds.
- The vowel /**a**/, close to the vowel in "Don," is modeled as a **2 Hz vibration plus a 3 Hz vibration**:

$$x(t) = \cos(4\pi t) + \cos(6\pi t)$$

- At $t = 0$, both cosines equal $+1$, so /a/ also starts at $x = +2$, which is maximum rarefaction.
- At $t = 0.5$ seconds, the 2 Hz wave has completed exactly one cycle and the 3 Hz wave has completed one and a half cycles, so the sum is:

$$x(0.5) = \cos(2\pi) + \cos(3\pi) = 1 + (-1) = 0$$

- The /a/ wave has a different shape from /i/. Between each large peak at $+2$, it drops to a deep trough near -1.6, rises to a pair of small humps near $+0.7$ separated by a dip down to zero, drops to another deep trough, and returns to $+2$.
- This pattern also repeats **once every second**, returning to $+2$ at 1, 2, and 3 seconds.
- In both vowels, **positive x means rarefaction and negative x means compression**. Both vowels start at the same value and repeat at the same rate, yet the air moves in a clearly different way within each second. **The combination of frequencies is what distinguishes one vowel from another.**
    - The motion in these models is played back at one quarter of its speed so that the particle spacing can be followed.

Section 12: Perturbation Theory

- **Perturbation theory** is the study of how a system's behavior changes when one of its parameters is nudged up or down. A **parameter** is a fixed setting of a system, such as how heavy the mass is or how stiff the spring is. For a mass-spring system, the behavior of interest is its **frequency**, meaning how many oscillations it completes per second.
- **If we increase the mass m of a mass-spring system, the frequency decreases.**
    - A heavier mass carries its motion more strongly, so the spring takes longer to slow it down and turn it around at each end of its swing.
    - Each cycle therefore takes longer, and fewer cycles fit into each second. **More mass means a lower frequency.**
- **If we increase the stiffness k of the spring, the frequency increases.**
    - A stiffer spring pulls back harder on the mass for the same displacement, so it turns the mass around more quickly at each end of its swing.
    - Each cycle therefore takes less time, and more cycles fit into each second. **More stiffness means a higher frequency.**
- These two results together form the core perturbation-theory relationships for a mass-spring system:
    - **Increasing mass lowers frequency.**
    - **Increasing stiffness raises frequency.**
    - These relationships give us a way to predict how a vibrating system will sound or behave just from knowing whether it is heavy or light and stiff or loose.

Section 13: Hearing Explained

- The mass-spring relationships explain how the ear separates sounds by frequency. This explanation comes from the work of **Georg von Békésy**, who received the **Nobel Prize in 1961** for discovering how the inner ear responds to sound.
- Inside the inner ear is a long, tapered membrane called the **basilar membrane**. It is wide at one end and narrow at the other, and its physical properties change gradually along its length.
    - At the **wide end**, the membrane has **large mass M** and **small stiffness k**. It is heavy and loose.
    - At the **narrow end**, the membrane has **small mass m** and **large stiffness K**. It is light and stiff.
    - Moving along the membrane from the wide end to the narrow end, mass steadily decreases and stiffness steadily increases.
- Applying perturbation theory to each location on the membrane gives each location its own preferred frequency.
    - At the wide end, large mass lowers the frequency and small stiffness also lowers the frequency. This end vibrates most strongly for **low-frequency** sounds.
    - At the narrow end, small mass raises the frequency and large stiffness also raises the frequency. This end vibrates most strongly for **high-frequency** sounds.
    - Every point in between has an intermediate mass and stiffness, so every point is tuned to an intermediate frequency. The membrane is effectively a long row of

mass-spring systems, each set to a slightly different frequency.

- **Nerve fibers** leave the membrane at many different locations along its length. When a particular spot on the membrane vibrates, the fibers at that spot become active, and which fibers are active tells the brain which frequencies are present in the sound.
    - When a vowel made of two combined frequencies reaches the ear, such as the 1 Hz and 4 Hz combination for /i/, two different places along the membrane vibrate, one for each frequency. The membrane separates the combined vibration back into its component frequencies.
    - This is how the ear can tell /i/ from /a/: the two vowels are built from different frequency combinations, so they set different places on the membrane into vibration.

Section 14: The Fundamental Learning Tradeoff

- In machine learning, a model is trained by repeatedly updating a value to move it toward a target. The **learning rate**, written γ, sets how large each update is, as a fraction of the remaining distance to the target. Each full round of updates through the training data is called an **epoch**.
- The effect of γ can be seen by training three runs that start at very different values, one near 130, one near 80, and one near 10, and all head toward a target of about 80. Each run is tracked over 300 epochs at four different learning rates.
    - At a **learning rate of 1%**, the runs move smoothly and slowly. The run starting at 130 curves down gradually and is still at about 82 after 300 epochs. The run starting at 10 curves up gradually and has only reached about 76 after 300 epochs. The motion is very steady, but the runs have not fully reached the target even by the end.
    - At a **learning rate of 10%**, the runs reach the neighborhood of 80 within roughly the first 30 to 50 epochs. After arriving, they stay close to 80 with small wiggles of a few units above and below.
    - At a **learning rate of 50%**, the runs jump to the region of 80 within just a few epochs. After that, they bounce around noisily, swinging roughly between 70 and 95 rather than sitting on the target.
    - At a **learning rate of 80%**, the runs reach the target region almost immediately, but they bounce even more widely, roughly between 68 and 97, for all 300 epochs.
- These results show the **fundamental learning tradeoff**:
    - **When γ increases, speed increases and accuracy decreases.** Large updates get the value to the target region quickly, but the same large updates keep throwing the value past the target, so it oscillates around the answer instead of settling on it.
    - **When γ decreases, speed decreases and accuracy increases.** Small updates keep the value on a smooth, steady path with very little bouncing, but they take many more epochs to arrive.
    - This is the **oscillation in SGD** that appears among the everyday examples of vibration. A high learning rate turns training into a back-and-forth swinging motion around the target.
- The tradeoff is an unfortunate one, since in practice we want both fast learning and accurate learning at the same time. **Thousands of people, over generations, have made their living by finding better ways to optimize**, meaning better training methods that raise both speed and accuracy together rather than trading one against the other.
    - The goal of this work is a method where **speed increases** and **accuracy increases** at once, escaping the tradeoff that a single fixed learning rate forces on us.

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Section 1: The Harmonic Oscillator

- A **harmonic oscillator** is a dynamical system built from two variables that drive each other: **position**, written x, and **velocity**, written v. The dot above a letter marks change over time, so $\dot{x}$ means the **increase or decrease in x with time**, and $\dot{v}$ means the **increase or decrease in v with time**. Each variable has its own rule of change, and each rule depends on the other variable.
- The first rule says that position changes at a rate equal to the velocity:

$$\dot{x} = v$$

- Velocity drives position. When v is positive, x grows, when v is negative, x shrinks, and when v is zero, x holds still for that instant.
- The second rule says that velocity changes at a rate equal to the negative of the position:

$$\dot{v} = -x$$

- Position drives velocity, but with a minus sign. A positive x pushes velocity downward, and a negative x pushes velocity upward.
- Together the two rules form a **closed loop of mutual influence**, where v changes x and x changes v over and over.
- Stepping through the loop by hand from $x = 1$ and $v = 0$ shows how it produces repeating motion.
    - With $x = 1$, the second rule gives $\dot{v} = -1$, so velocity becomes negative.
    - Once v is negative, the first rule makes $\dot{x}$ negative, so position starts to fall. It falls slowly at first and then faster as v grows more negative.
    - When position reaches $x = 0$, the second rule gives $\dot{v} = 0$, so velocity stops changing. Velocity is at its most negative at this moment, which means position is falling fastest as it passes through zero.
    - Once x is negative, $\dot{v}$ is positive, so velocity climbs back toward zero and position falls more slowly.
    - When velocity reaches zero, position stops falling at its lowest point, -1. Velocity then turns positive, position rises, and the whole sequence runs in reverse until x is back at $+1$ with $v = 0$.
- Plotted against time, both variables trace waves. Position starts at its peak of $+1$, drops through zero to -1, and returns to its peak. Velocity starts at zero, dips negative, returns to zero, rises positive, and returns to zero.
    - The two waves are offset by a quarter of a cycle. **Velocity is zero whenever position is at a peak or trough, and velocity is largest whenever position crosses zero.**
    - This offset follows directly from the two rules, since position turns around only when velocity reaches zero, and velocity turns around only when position reaches zero.

Section 2: The Oscillation/Vibration Dynamical System

- An **oscillation**, also called a **vibration**, is a repeating back-and-forth motion around a central point. The two-variable loop between x and v is the dynamical system that produces it. The influence runs in a circle, with x acting on v and v acting back on x, so there is no single starting cause and no single final effect.
    - Over time, x traces a smooth wave that starts high, swings below zero, swings back up, and repeats. v traces a wave of the same shape that starts at zero, heads negative first, and stays a quarter cycle out of step with x.

- **This system never comes to rest on its own.** Every time one variable reaches zero, the other variable is at its largest value and pushes it past zero again.
    - Position overshoots zero since velocity is at its maximum at the moment position reaches zero.
    - Velocity overshoots zero since position is at its maximum at the moment velocity reaches zero.

Section 3: Vibration Needs Two Physical Effects

- Vibration requires **two physical effects working together**.
- The first effect is a **viscous spring**, which is the springiness found in many things around us, meaning their tendency to return to a natural position after being moved. The word "viscous" refers to a frictional, slowing quality and is set aside here. A spring has a property called **stiffness**, which measures how strongly it pulls back toward its natural position.
- The second effect is **mass**, which determines how an object carries motion once it is moving.
- Vibration appears across many different systems.
    - It is **what we feel when a car hits a bump**.
    - It is **music**.
    - It is **boom and bust in economics**, where an economy swings above and below a steady level in repeating cycles.
    - It is **oscillation in SGD**. SGD stands for **stochastic gradient descent**, a method for training machine-learning models by repeatedly updating a value toward a target, and under some settings that value swings back and forth around the target.
    - It is **the motion of the air in the vocal tract and the motion of the vocal folds during speech**.

Section 4: A Slice of Air in the Vocal Tract

- A slice of air in the vocal tract is made of many particles. During vibration, the particles stay the same and each one stays the same size, but **the distances between them change**.
    - At **equilibrium spacing**, the particles sit at their resting distance. A chosen pair of neighboring particles is at **100% of its equilibrium distance**.
    - In **compression**, the particles are pushed closer together, and the chosen pair shrinks to **56% of its equilibrium distance. High pressure means air packed very close together**, so compression is a high-pressure state.
    - In **rarefaction**, the particles spread farther apart, and the chosen pair stretches to **144% of its equilibrium distance. Low pressure means air spread very far apart**, so rarefaction is a low-pressure state.
- A vibrating slice of air alternates between compression and rarefaction, shrinking and expanding around its equilibrium size. In this idealized model, the slice's equilibrium size serves as the reference, and the motion is exaggerated so the change in spacing is visible.

Section 5: The Viscous Spring and Negative/Balancing Feedback

- The spring effect alone is described by one rule of change:

$$\dot{x} = -\frac{1}{3}x$$

- Here x is the distance from the spring's natural position, and $\dot{x}$ is how fast that distance changes. The rate of change is always **one third of the current distance, with the opposite sign**.
- The sign of x decides the direction of change.
    - **If x is positive, $\dot{x}$ is negative, so x falls** toward zero.
    - **If x is negative, $\dot{x}$ is positive, so x rises** toward zero.
- Starting from an **initial condition** of $x = 1$, meaning the value the system begins at, x falls by one third of its current size. One third of 1 is $\frac{1}{3}$, so x becomes:

$$1 - \frac{1}{3} = \frac{2}{3}$$

- The rule then applies again to the new value. One third of $\frac{2}{3}$ is $\frac{2}{9}$, so x becomes:

$$\frac{2}{3} - \frac{2}{9} = \frac{4}{9}$$

- Applying it once more, one third of $\frac{4}{9}$ is $\frac{4}{27}$, so x becomes:

$$\frac{4}{9} - \frac{4}{27} = \frac{8}{27}$$

- This repeats again and again, with each drop smaller than the last. **When x is smaller, $\dot{x}$, its speed of falling, is smaller too.**
- Graphed over time, x starts at 1 and curves down toward zero. The slope starts steeply negative and becomes **smaller and smaller negative**, so the curve flattens as it nears zero and never crosses it. With a negative initial condition, x rises toward zero from below with a slope that becomes smaller and smaller positive, again approaching zero without crossing it.
- As in Meadows' account of systems, these systems have a **natural resting condition**. When their position is at a **discrepancy** with that natural position, they **feed back** on themselves through $\dot{x} = -\frac{1}{3}x$ to cancel the discrepancy. This is **negative feedback**, also called **balancing feedback**, since every change opposes the discrepancy and brings the system back into balance.
    - **These systems are in motion only in order to achieve their equilibrium.** An **equilibrium** is a state where the system has no reason to change further, so once the spring arrives at its natural position, its motion stops.
- The number $-\frac{1}{3}$ is the **stiffness**. A higher stiffness closes a larger share of the remaining distance at each step, so the spring returns to equilibrium faster. A lower stiffness closes a smaller share at each step, so the return is slower.

Section 6: Mass and Persistent Motion

- **If a mass is moving, it just keeps on moving.** A mass has no natural resting position that it is trying to reach and needs no discrepancy to keep going.
- **Galileo, Descartes, and Newton** established that **a mass in motion stays in motion**. A mass moving at **constant velocity** covers **the same distance in every equal interval of time**, without speeding up, slowing down, or stopping on its own. A spring slows as it nears its resting point and stops there, while a mass carries its motion forward regardless of where it is.

Section 7: A Spring Stops at a Place, a Mass Stops After a Process

- **A spring stops at a place, while a mass stops after a process.** The spring's stopping point is a location, its natural position, while a moving mass continues until its motion is removed over time.
- For a spring attached to a wall with **no mass**, displacement x is **measured from the spring's natural position** at $x = 0$, on a scale from -1 to $+1$.
    - Pulled to $+1$ and released, the spring **returns and stops at** $x = 0$, approaching zero **without crossing it**. Over time, this return is a curve that bends down from $+1$ and flattens as it reaches zero.
    - Pushed to -1 and released, the spring **rises toward zero without crossing it**, following a curve that bends up and flattens.
    - Repeating the pull gives the same return every time, since a spring alone always settles at its natural position.
- With a **mass** attached and the system released from $+1$, the spring pulls the mass toward zero, but the mass is moving fastest when it reaches zero and **carries its motion through the crossing.**
    - The mass continues into negative displacement, where the spring pulls it back toward zero from the other side. It slows, stops near -1, and is pulled back through zero toward the positive side.

    - **The mass carries motion through every crossing**, so it keeps swinging back and forth past the natural position.
- Over time, the spring-plus-mass system produces a **lightly damped oscillation**, a curve that swings from $+1$ below zero, back above zero, and on through repeated cycles. **Damping** is the gradual loss of motion from the viscous part of the spring. When damping is light, each swing is only slightly smaller than the one before, so the mass stops only after a long process of swings.
- The two effects together form the harmonic oscillator. The spring's pull toward the natural position corresponds to $\dot{v} = -x$, and the mass's persistent motion corresponds to $\dot{x} = v$.
    - A spring alone gives balancing feedback that ends at equilibrium.
    - A mass alone gives motion that never turns around.
    - **A spring and a mass together give vibration**, since the spring keeps turning the motion around and the mass keeps carrying it past the resting point.

Section 8: A Cosine in the Vocal Tract

- The vibration of a slice of air in the vocal tract follows a **cosine**. Here x measures the **normalized expansion** of the slice, scaled so its extremes are $+1$ and -1:

$$x = \cos(\text{phase})$$

- $x = +1$ means the particles are **farthest apart**, which is **maximum rarefaction**.
- $x = 0$ means the particles are at **equilibrium** spacing.
- $x = -1$ means the particles are **closest together**, which is **maximum compression**.
- The **phase** tracks where the slice is within its cycle, and one full cycle takes a phase of 2π. A phase from 0 to 6π therefore covers **three cycles**.
    - At phase 0, $x = +1$, so the slice starts at maximum rarefaction.
    - At phase π, $x = -1$, so the slice reaches maximum compression halfway through Cycle 1.
    - At phase 2π, x returns to $+1$, completing Cycle 1. Cycle 2 ends at 4π and Cycle 3 ends at 6π.
- In every cycle, the slice moves from **maximum rarefaction to compression and back to rarefaction.** In between the extremes, $x = +0.84$ is a rarefied state with particles farther apart than equilibrium but not at their widest, and $x = -0.84$ is a compressed state with particles closer than equilibrium but not at their tightest.
    - This cosine has the same shape as the position wave of the harmonic oscillator, starting at its peak and swinging through zero to its lowest point and back.

Section 9: Vowels and Tongue Position

- Vowels differ in where the tongue body sits inside the vocal tract, along two dimensions. **Height** is how high or low the tongue body sits, and **front/back** is how far forward or backward it sits.
- **High vowels** have the tongue body raised toward the roof of the mouth.
    - In **"dean,"** the vowel is **high and front**. The tongue body is raised and pushed forward, leaving a wide space behind the tongue.
    - In **"dune,"** the vowel is **high and back**. The tongue body is raised and bunched toward the back of the mouth.
- **Low vowels** have the tongue body lowered away from the roof of the mouth.
    - In **"Dan,"** the vowel is **low and front**. The tongue sits low while staying forward, leaving a large open space above it.
    - In **"Don,"** the vowel is **low and back**. The tongue is lowered and pulled back, narrowing the space behind it.
- In **"done,"** the vowel is **central**, with the tongue body in a middle position that is neither far forward nor far back.

Section 10: One Time Interval, Three Frequencies

- **Frequency** is the number of complete cycles a vibration goes through in a given time. Three vibrations can be compared over the same **12-second interval**, each covering the same range of particle spacing and each starting at **maximum rarefaction**, $x = +1$.
- The **low-frequency** vibration completes **3 cycles in 12 seconds**, so each cycle takes 4 seconds and its peaks fall at 0, 4, 8, and 12 seconds:

$$\frac{3 \text{ cycles}}{12 \text{ s}} = 0.25 \text{ cycles per second}$$

- The **mid-frequency** vibration completes **6 cycles in 12 seconds**, so each cycle takes 2 seconds:

$$\frac{6 \text{ cycles}}{12 \text{ s}} = 0.5 \text{ cycles per second}$$

- The **high-frequency** vibration completes **9 cycles in 12 seconds**, so each cycle takes $\frac{12}{9}$, or about 1.33 seconds:

$$\frac{9 \text{ cycles}}{12 \text{ s}} = 0.75 \text{ cycles per second}$$

- All three reach the same extremes of rarefaction and compression. **The only difference is how quickly the cycle repeats**, so frequency is separate from how far the particles spread or squeeze. One cycle per second is **one hertz**, written **Hz**.

Section 11: Vowels as Combined Vibrations

- A vowel can be modeled as **two vibrations at different frequencies added together** in the same slice of air, with equal component amplitudes. These are illustrative two-frequency models, and the pair of frequencies gives each vowel its own pattern of motion.
  - A cosine at f Hz can be written as $\cos(2\pi f t)$, where t is time in seconds. One cycle takes a phase of 2π, so $\cos(2\pi t)$ completes one cycle per second and $\cos(8\pi t)$ completes four.
- The vowel /**i**/ is modeled as **1 Hz + 4 Hz**:

$$x(t) = \cos(2\pi t) + \cos(8\pi t)$$

- At $t = 0$, both cosines equal $+1$:

$$x(0) = 1 + 1 = 2$$

- The combined wave starts at $+2$, which is **maximum rarefaction**, and its range runs from $+2$ to -2.
- At $t = 0.5$ seconds:

$$x(0.5) = \cos(\pi) + \cos(4\pi) = -1 + 1 = 0$$

- Within each second, the fast 4 Hz wave rides on the slow 1 Hz wave. The /i/ wave drops from $+2$ to troughs near -1.7, rises to smaller peaks near $+1$, touches zero at the halfway point, and returns to $+2$ at 1, 2, and 3 seconds.
- The vowel /**a**/ is modeled as **2 Hz + 3 Hz**:

$$x(t) = \cos(4\pi t) + \cos(6\pi t)$$

- At $t = 0$, both cosines equal $+1$, so /a/ also starts at $x = +2$, maximum rarefaction.
- At $t = 0.5$ seconds:

$$x(0.5) = \cos(2\pi) + \cos(3\pi) = 1 + (-1) = 0$$

- Within each second, the /a/ wave drops from $+2$ to a deep trough near -1.6, rises to two small humps near $+0.7$ separated by a dip to zero, drops to another deep trough, and returns to $+2$ at 1, 2, and 3 seconds.
- In both models, **positive x means rarefaction and negative x means compression**, and the motion is played back at **one quarter speed**. Both vowels start at $+2$ and repeat once

per second, but the air moves in a different pattern within each second, and **that pattern is set by the combination of frequencies**.

Section 12: Perturbation Theory

- **Perturbation theory** asks how a system's behavior changes when one of its settings is increased or decreased. For a mass-spring system, the behavior is its **frequency**.
- **If the mass m increases, the frequency decreases.** A heavier mass carries its motion more strongly, so the spring takes longer to turn it around at each end of its swing, and fewer cycles fit into each second.
- **If the stiffness k increases, the frequency increases.** A stiffer spring pulls back harder at the same displacement, so it turns the mass around more quickly, and more cycles fit into each second.
- The two perturbation relationships are that **increasing mass lowers frequency** and **increasing stiffness raises frequency**.

Section 13: Hearing Explained

- **Georg von Békésy** received the **Nobel Prize in 1961** for explaining hearing using these mass and stiffness relationships.
- Hearing depends on a long, tapered membrane in the ear whose mass and stiffness change gradually along its length.
  - At the **wide end**, the membrane has **large mass** M and **small stiffness** k, so it is heavy and loose.
  - At the **narrow end**, it has **small mass** m and **large stiffness** K, so it is light and stiff.
  - Moving from the wide end to the narrow end, mass steadily decreases and stiffness steadily increases.
- Applying perturbation theory gives each location its own frequency.
  - At the wide end, large mass and small stiffness both lower the frequency, so this end vibrates for **low frequencies**.
  - At the narrow end, small mass and large stiffness both raise the frequency, so this end vibrates for **high frequencies**.
  - Each point in between has intermediate mass and stiffness, so it vibrates for an intermediate frequency.
- Fibers leave the membrane at many points along its length. The location that vibrates determines which fibers carry a signal, so **the place of vibration along the membrane encodes the frequency of the sound**.

Section 14: The Fundamental Learning Tradeoff

- In training, a value is updated repeatedly toward a target, and the **learning rate**, written γ, sets how large each update is. Each full round of training updates is an **epoch**.
- Three runs starting near 130, near 80, and near 10 all move toward a target of about 80 over **300 epochs**, and the learning rate changes how they get there.
  - At a **learning rate of 1%**, the runs move smoothly and slowly. After 300 epochs, the run from 130 is still near 82 and the run from 10 has reached only about 76.
  - At a **learning rate of 10%**, the runs reach the region of 80 within roughly the first 30 to 50 epochs and then stay near it with small wiggles.
  - At a **learning rate of 50%**, the runs reach 80 within a few epochs but then bounce noisily, roughly between 70 and 95.
  - At a **learning rate of 80%**, the runs reach the target region almost immediately but bounce even more widely, roughly between 68 and 97, for all 300 epochs.
- These results define the **fundamental learning tradeoff**.
  - **When γ increases, speed increases and accuracy decreases.** Large updates reach the target region quickly, but they keep overshooting, so the value oscillates around the target instead of settling on it. This is the **oscillation in SGD**.
  - **When γ decreases, speed decreases and accuracy increases.** Small updates follow a smooth, steady path with little bouncing, but they need many more

epochs to arrive.

- The tradeoff is unwelcome, since fast and accurate learning are both wanted at once. **Thousands of people for generations have made their living by finding better ways to optimize**, so that **speed increases** and **accuracy increases** together.